

DESIGN, SIMULATION AND VERIFICATION OF A
520 kW GRID-CONNECTED PV MICROGRID
WITH MPPT CONTROL AND A TIME-OF-USE
BATTERY ENERGY MANAGEMENT SYSTEM

A MATLAB / Simulink Simscape Electrical Study

Muhammad Umer Mujahid

ACM Group of Industries

umermujahidofficial@gmail.com

2026

Keywords: Photovoltaic Systems · MPPT · Boost Converter · Voltage Source Inverter · Grid Synchronization · Battery Energy Storage · Time-of-Use Tariff · Simulink

Table of Contents

Abstract

Glossary of Terms and Abbreviations

I. Introduction

II. System Design and Component Working Principles

III. Results and Findings — PV Plant

IV. Energy Management System — Design and Results

V. Discussion

VI. Future Work

VII. Conclusion

References

Appendix A — Instructions for Inserting Your Own Circuit Diagrams and Graphs

Abstract

This paper presents the design, simulation, and rigorous multi-stage verification of a 520 kW-class grid-tied photovoltaic (PV) microgrid, developed in MATLAB/Simulink using the Simscape Electrical Specialized Power Systems library. The plant comprises three independently designed PV-array, boost-converter, and Perturb-and-Observe (P&O) Maximum Power Point Tracking (MPPT) chains, each feeding a three-phase two-level Voltage Source Converter (VSC) inverter through an LC filter and an isolated, grounded-wye step-up transformer onto a common AC bus. Rather than accepting simulation output at face value, every subsystem in this study was checked against closed-form power-electronics theory or cross-validated by an independent measurement method: continuous-conduction-mode (CCM) boost-converter behaviour was confirmed to within 2.7% of the ideal $V_{out} = V_{in}/(1-D)$ relation; MPPT convergence was verified through a controlled step-change in solar irradiance; and AC-side active power delivery was independently cross-checked using two measurement methods, converging on a combined DC generation capacity of approximately 520 kW and a validated AC delivery of approximately 406 kW ($\approx 77\%$ conversion efficiency, near-unity power factor). Building on this validated plant, a companion 24-hour Energy Management System (EMS) was developed, incorporating a Pakistan-specific solar generation profile, a residential load profile, a Time-of-Use (ToU) tariff structure, a battery state-of-charge (SOC) tracker, and a rule-based scheduling controller governing three flexible household loads. Comparative simulation against an unscheduled baseline demonstrated an approximately 3% reduction in net daily electricity cost under smart, tariff-aware scheduling.

Index Terms — Photovoltaic systems, Maximum Power Point Tracking, boost converter, voltage source inverter, grid synchronization, battery energy storage, demand-side management, Time-of-Use tariff, Simulink, Simscape Electrical.

Glossary of Terms and Abbreviations

This glossary defines every technical term and abbreviation used in this paper, for readers who may not be familiar with power-electronics terminology.

Term	Meaning
PV	Photovoltaic — a device or system that converts sunlight directly into electricity.
MPPT	Maximum Power Point Tracking — a control technique that continuously adjusts a PV array's operating point to extract the maximum power available at any given sunlight and temperature condition.
P&O	Perturb and Observe — a specific, widely-used MPPT algorithm that makes a small change ("perturbation") to the operating point and observes whether power increased or decreased, then repeats.
Duty cycle (D)	The fraction of each switching period that a power-electronic switch (e.g. an IGBT) spends in the ON state. Expressed as a value between 0 and 1.
IGBT	Insulated Gate Bipolar Transistor — a semiconductor switch used in power electronics to rapidly turn current on and off, forming the building block of converters and inverters.
PWM	Pulse Width Modulation — a technique for controlling power-electronic switches by varying the width (ON time) of switching pulses to synthesize a desired average voltage or current.
Boost converter	A DC-DC converter topology that steps a lower input DC voltage up to a higher, regulated output DC voltage.
CCM	Continuous Conduction Mode — a boost-converter operating condition in which current through the inductor never falls to zero during a switching cycle.
DCM	Discontinuous Conduction Mode — a condition in which inductor current falls to zero before the next switching cycle begins, causing the converter's behaviour to deviate from the standard CCM equations.
VSC	Voltage Source Converter — a type of inverter that uses switching devices to synthesize an AC voltage waveform from a DC source.
Inverter	A power-electronic circuit that converts DC electricity into AC electricity.
LC filter	A filter built from an inductor (L) and capacitor (C) used to smooth the high-frequency switching harmonics out of an inverter's raw output, leaving a cleaner sinusoidal waveform.
Transformer	An electrical device that transfers energy between two circuits at different voltage levels using magnetic coupling, and can also provide electrical isolation between them.
Yg (grounded-wye)	A three-phase winding connection in which the neutral point is both accessible and connected to ground, providing a stable reference for voltage measurement.
RMS	Root Mean Square — a mathematical way of expressing the effective, equivalent magnitude of a continuously varying (AC) signal.
Active power (P)	The real, usable power delivered to a load, measured in watts (W) — the power that actually does work.
Reactive power (Q)	Power that oscillates between source and load without doing net work, measured in volt-amperes-reactive (var); associated with inductive or capacitive elements.

Apparent power (S)	The combination of active and reactive power, $S = \sqrt{P^2+Q^2}$, measured in volt-amperes (VA).
Power factor (PF)	The ratio of active power to apparent power, $PF = P/S$; a PF close to 1.0 ("unity") indicates efficient, mostly-real power delivery.
PLL	Phase-Locked Loop — a control system that locks an internal reference signal's frequency and phase to match an external reference, used to synchronize an inverter with the utility grid before connection.
SOC	State of Charge — the remaining energy in a battery, expressed as a percentage of its total capacity.
EMS	Energy Management System — a control layer that decides how to charge/discharge storage and schedule loads to minimize cost or meet other objectives.
ToU tariff	Time-of-Use tariff — an electricity pricing structure in which the price per unit of energy varies depending on the time of day (e.g. cheaper off-peak, more expensive during peak demand hours).
kW / kWh	Kilowatt (unit of power, 1,000 watts) / Kilowatt-hour (unit of energy — power sustained for one hour).
PKR	Pakistani Rupee, the currency used for the tariff and cost figures in the EMS study.

I. Introduction

A. Background and Motivation

Grid-connected photovoltaic systems are a core building block of modern low-carbon power infrastructure. Reliable operation depends on the correct interaction of several power-electronic subsystems in series: the PV array itself, a DC-DC boost stage with maximum power point tracking, a DC-AC inverter, output filtering, and a grid interface providing correct voltage transformation and reference isolation. A weakness at any single stage — an under-damped converter, a stalled tracking algorithm, an unsynchronized grid connection — silently degrades the performance of every stage downstream of it.

B. Project Objectives

- Design a realistic, datasheet-grounded 520 kW-class PV plant using three parallel inverter chains, matching a genuine utility-scale string-and-inverter layout.
- Implement Maximum Power Point Tracking from first principles (original algorithm code, not a pre-built library block) and verify it responds correctly to changing sunlight conditions.
- Verify every subsystem's behaviour against closed-form power-electronics theory or an independent measurement method, rather than accepting simulation output at face value.
- Extend the validated plant with a 24-hour battery-storage and demand-scheduling Energy Management System, evaluated against a realistic Pakistani residential electricity tariff.

C. Scope of This Work

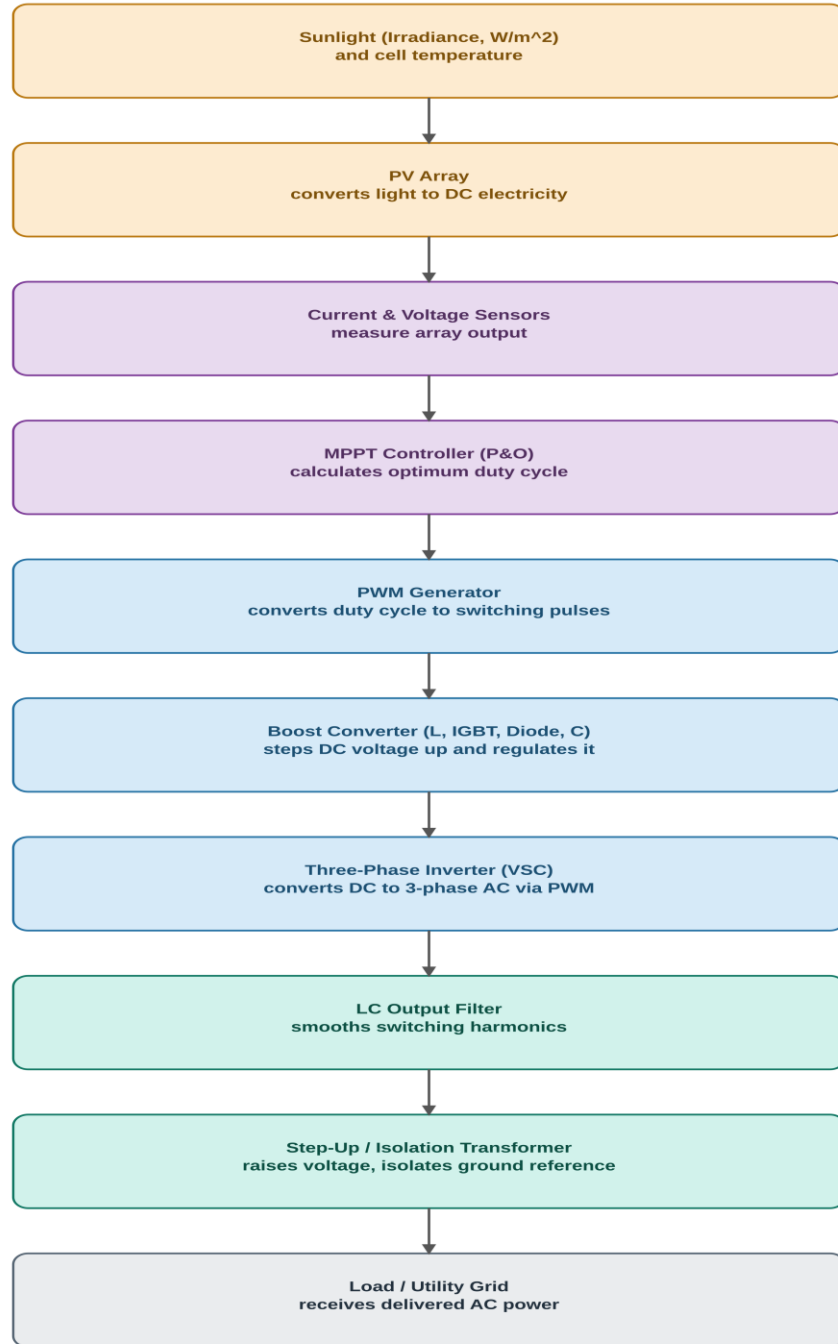
This work builds each subsystem from first principles in Simulink's Specialized Power Systems (Simscape Electrical) environment, using manufacturer datasheet parameters rather than idealized library defaults, and subjects every subsystem to independent numerical verification before integrating it with the next. This verification-first methodology is the central contribution of the present study: each design decision reported in Section III is accompanied by the evidence used to accept or reject it, including two cases (a discontinuous-conduction fault and a grid-synchronization fault) that were deliberately encountered, diagnosed, and corrected rather than avoided by construction.

The remainder of this paper is organized as follows. Section II describes every subsystem's working principle, design, and role in the overall plant. Section III presents simulation results and the verification evidence for the PV plant. Section IV covers the EMS design and its results. Section V discusses the findings and their root causes, Section VI outlines future work, and Section VII concludes.

II. System Design and Component Working Principles

This section explains every major component in the system, what physical role it plays, and how it was designed. Fig. 1 shows the overall signal flow from sunlight to the grid, and each subsystem is then described in the order of energy passes through it.

Overall Signal Flow — Sunlight to Grid



Note: the MPPT controller and boost converter form a closed control loop, continuously re-measuring V and I every switching cycle.

Fig. 1. Overall signal flow of a single inverter chain, from sunlight to the AC grid/load.

A. Plant-Level Architecture

Each inverter chain follows the path shown in Fig. 1 above; three such chains are combined in parallel on a common AC bus to reach the target plant capacity (Fig. 2), matching a realistic utility-scale layout with mismatched string groups per inverter.

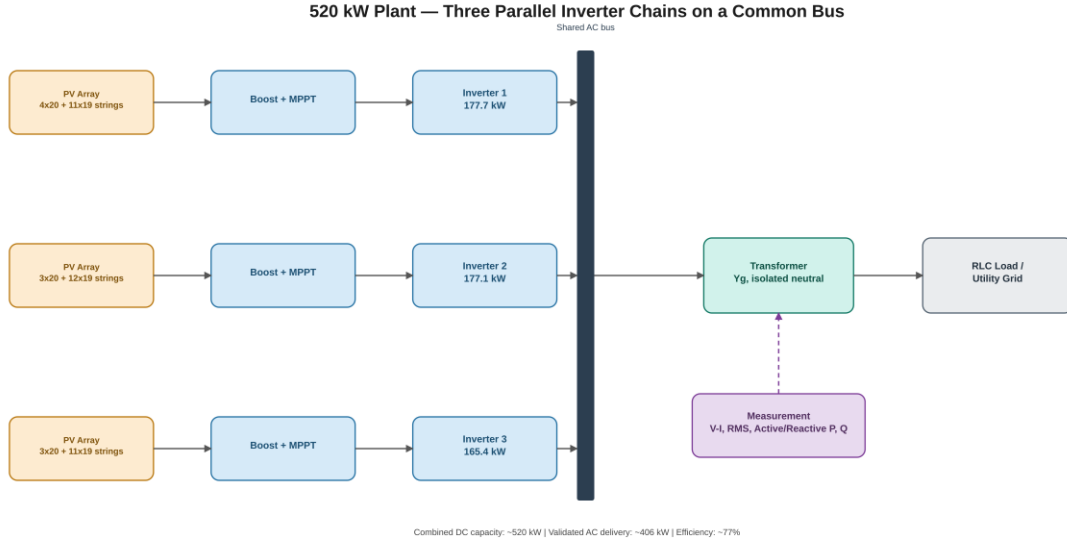


Fig. 2. Full plant architecture — three parallel inverter chains combining a common AC bus before the transformer and grid/load.

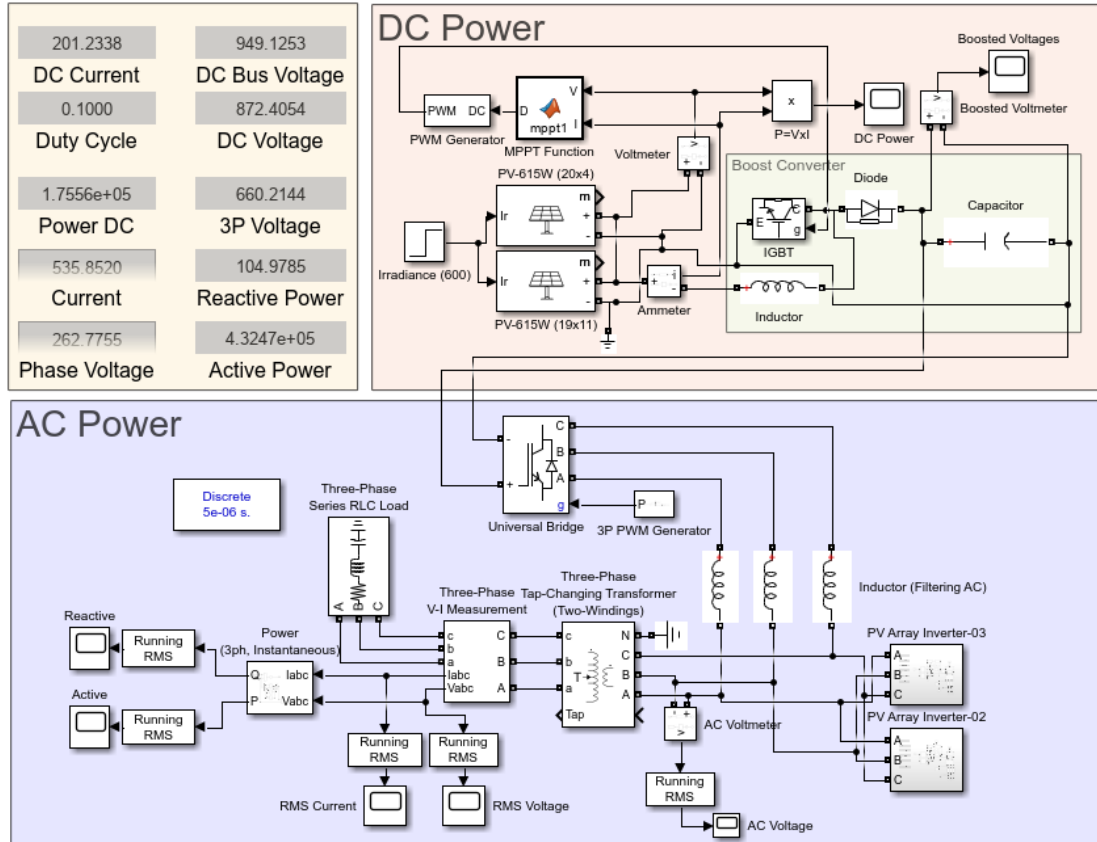


Fig. 3. Complete Simulink implementation of the 520 kW plant (all three chains).

#	Component	Role in the System
1	PV Array	Converts sunlight (irradiance) into DC electricity.
2	Voltage / Current Sensors	Measure the array's live operating voltage and current, feeding the MPPT controller.
3	MPPT Controller (P&O)	Calculates the duty cycle that keeps the array operating at its maximum power point.
4	PWM Generator	Converts the duty cycle number into actual switching pulses for the boost converter's IGBT.
5	Boost Converter (L, IGBT, Diode, C)	Steps the PV array's DC voltage up to a higher, regulated DC bus voltage.
6	Three-Phase Inverter (VSC)	Converts the regulated DC bus into three-phase AC power using high-frequency PWM switching.
7	LC Output Filter	Smooths the inverter's raw switching waveform into a clean sinusoid.
8	Step-Up / Isolation Transformer	Raises the AC voltage to the required level and provides an isolated, grounded neutral reference.
9	Grid / Load	Receives the delivered AC power.
10	Battery + EMS Controller	(Section IV) Stores surplus energy and schedules flexible loads to minimize cost.

Table I. Summary of every major component and its role in the system.

B. PV Array — Working Principle and Design

A PV array is built from individual solar modules, each of which converts sunlight into DC electricity through the photovoltaic effect: photons striking a semiconductor p-n junction excite electrons, producing a current when the circuit is closed. A single module's electrical behaviour is fully described by its I-V (current-voltage) and P-V (power-voltage) characteristic curves, which depend on irradiance (sunlight intensity, in W/m^2) and cell temperature.

Four datasheet quantities define these curves (see Glossary for full definitions): open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), and the voltage and current at the maximum power point (V_{mp} , I_{mp}). Multiplying $V_{mp} \times I_{mp}$ gives the module's rated power. As shown in Fig. 4 below, the P-V curve rises, peaks at a single point, and falls — this peak is precisely what the MPPT controller (Section II-D) is designed to find and track.

Each PV subsystem in this plant uses two parallel PV Array blocks, representing mixed 20-module and 19-module string groups, parameterized with manufacturer datasheet values for the Jinko Solar Tiger Neo N-type 78HL4 615 W module [1] ($V_{oc} = 56.25 \text{ V}$, $I_{sc} = 13.8 \text{ A}$, $V_{mp} = 46.81 \text{ V}$, $I_{mp} = 13.14 \text{ A}$, 156 half-cut cells), rather than generic library defaults.

Inverter	Strings (20-module / 19-module)	Design Target (W)
1	4 / 11	177,735
2	3 / 12	177,120
3	3 / 11	165,435
Total	44 strings	520,290

Table II. String allocation and design power target per inverter.

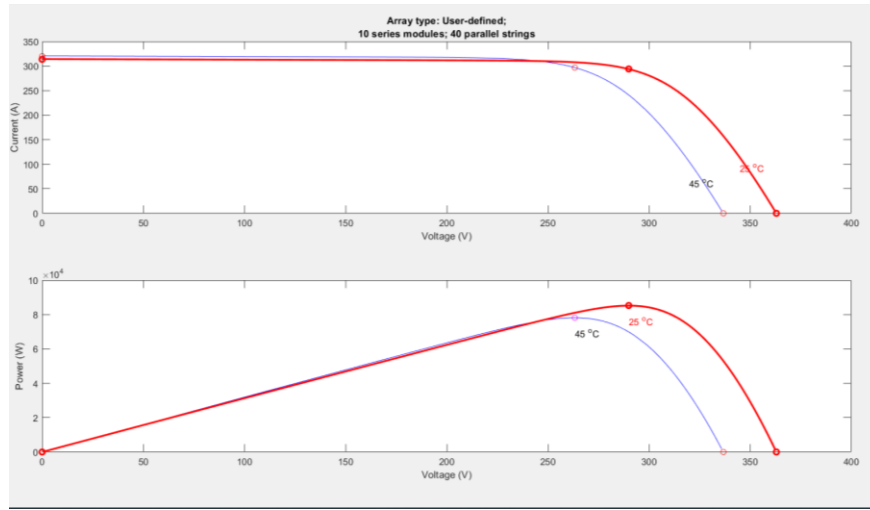


Fig. 4. Simulated I-V and P-V characteristics of the modeled PV array at 25°C and 45°C, 1000 W/m² (direct MATLAB output).



Fig. 5. PV array terminal voltage and current — sensor readings taken before the boost converter stage.

C. DC-DC Boost Converter — Working Principle and Design

The boost converter's job is to take the PV array's naturally low, sunlight-dependent DC voltage and step it up to a stable, higher DC bus voltage suitable for the inverter. It is built from four passive and active components, each with a distinct role:

- Inductor (L): stores energy in a magnetic field while the switch is ON, and releases that stored energy — at a boosted voltage — when the switch turns OFF. This is the physical mechanism that produces the voltage step-up.
- IGBT switch: rapidly turns ON and OFF (thousands of times per second) under PWM control, controlling how much energy the inductor stores each cycle. The fraction of each cycle spent ON is the duty cycle, D .
- Diode: provides a one-way path for current to flow from the inductor to the output only when the switch is OFF, preventing the output capacitor from discharging back through the switch.
- Output capacitor (C): smooths the diode's pulsed output current into a stable, low-ripple DC voltage.

In the boost converter's normal (continuous conduction mode, CCM) operating regime, the relationship between input and output voltage is fixed by the duty cycle alone:

$$V_{out} = V_{in} / (1 - D)$$

This closed-form relation was used throughout the study as the primary correctness check on every boost-converter operating point (see Section III-A).

Parameter	Value
Inductance (L)	5 mH
Switching frequency	5 kHz
IGBT snubber resistance / capacitance	100 k Ω / 1 μ F
Output capacitance	Sized for stable DC bus ripple
Duty cycle range (safety clamp)	0.1 – 0.9

Table III. Boost converter design parameters.

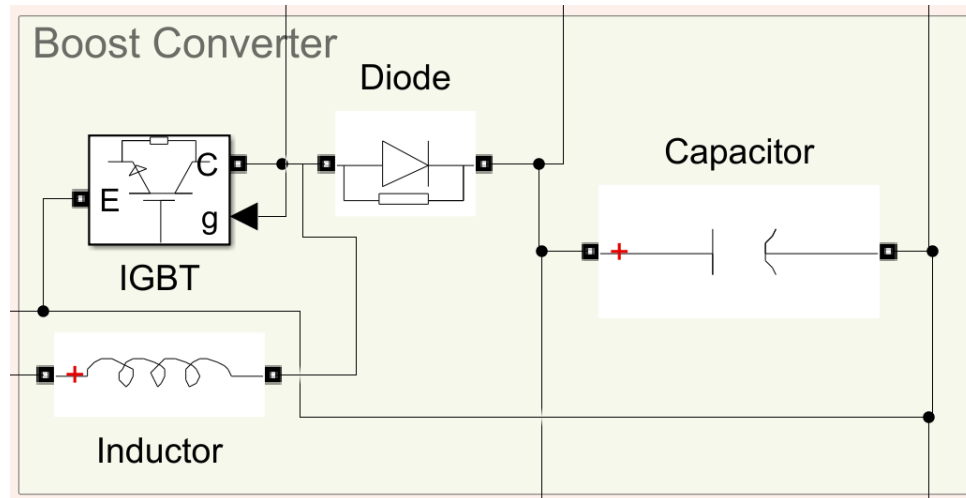


Fig. 6. Boost converter circuit implementation — inductor, IGBT, diode, and output capacitor.

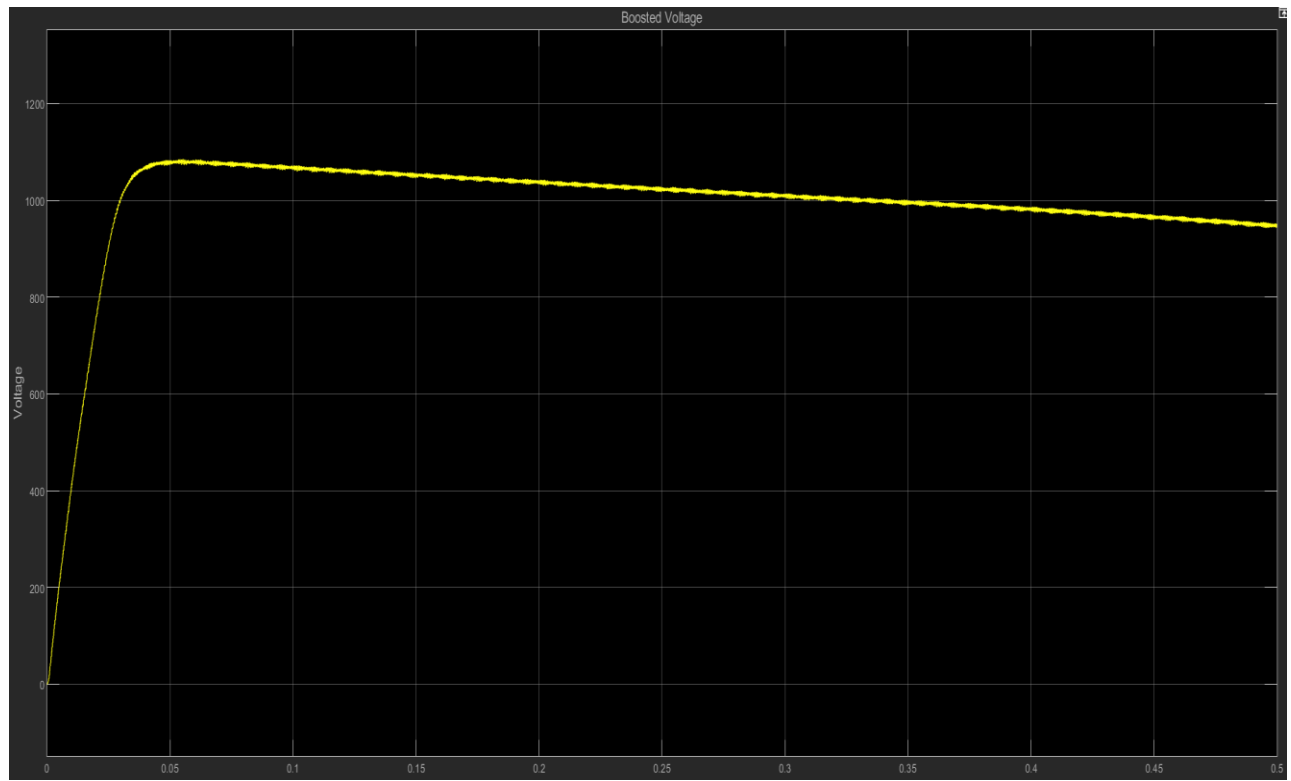


Fig. 7. DC bus voltage after the boost converter — regulated output feeding the inverter.

D. Maximum Power Point Tracking (MPPT)

A fixed duty cycle has no way of knowing where a PV array's true maximum-power operating point lies, and that point shifts continuously as sunlight and temperature change. MPPT solves this by continuously adjusting the boost converter's duty cycle to hunt for and hold the array at its instantaneous power peak.

This project implements the Perturb-and-Observe (P&O) algorithm [2] — one of the most widely used MPPT techniques in industry — as original MATLAB code (Algorithm 1) rather than a pre-built library block. On every control step, the algorithm measures the array's voltage and current, computes power, and compares it against the previous step: if power increased, the last adjustment direction is repeated; if power decreased, the direction is reversed. This simple rule causes the operating point to continuously "hill-climb" toward the true power peak. Fig. 8 below shows this logic as a flowchart.

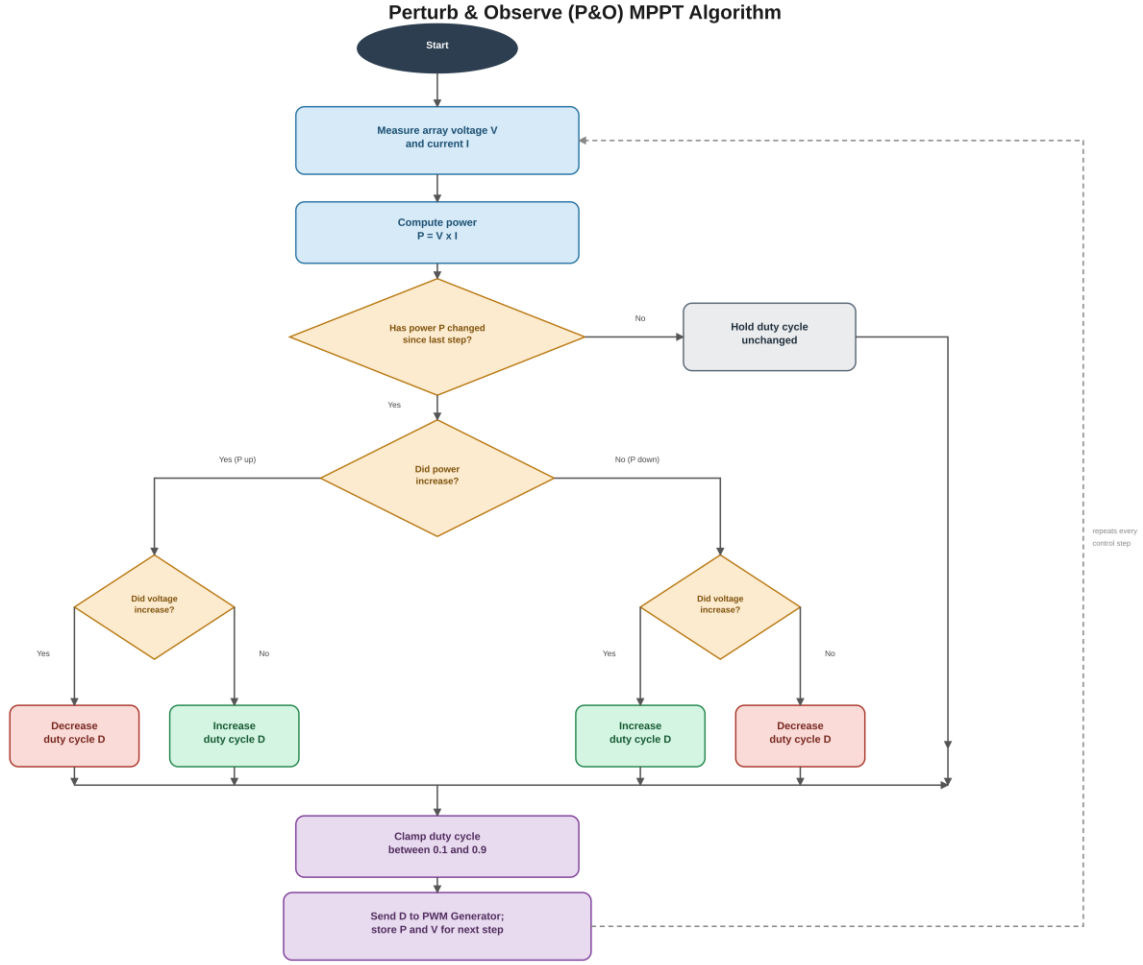


Fig. 8. The flowchart of the Perturb & Observe (P&O) MPPT algorithm implemented in this project.

Algorithm 1. Perturb-and-Observe MPPT (MATLAB Function block)

```

function D = mppt(V, I)
persistent Dprev Pprev Vprev
if isempty(Dprev)
    Dprev = 0.5; Pprev = 0; Vprev = 0;
end
P = V * I; dD = 0.0001;
if (P - Pprev) ~= 0
    if (P - Pprev) > 0
        if (V - Vprev) > 0, Dprev = Dprev - dD;
        else, Dprev = Dprev + dD; end
    else
        if (V - Vprev) > 0, Dprev = Dprev + dD;
        else, Dprev = Dprev - dD; end
    end
end
if Dprev >= 0.9, Dprev = 0.9;
elseif Dprev <= 0.1, Dprev = 0.1; end
D = Dprev; Pprev = P; Vprev = V;
end
  
```

E. Three-Phase Inverter (Voltage Source Converter)

The inverter's role is to convert the boost converter's regulated DC bus into three-phase AC power. This plant uses a two-level Universal Bridge — six IGBT/diode switch pairs, one pair per phase leg — switched at 2 kHz using a sinusoidal PWM generator. By rapidly switching the DC bus on and off in a pattern that approximates a sine wave, and then filtering the result (Section II-F), the bridge synthesizes a genuine three-phase AC output whose magnitude is controlled by the modulation index (the fraction of the DC bus voltage actually used).

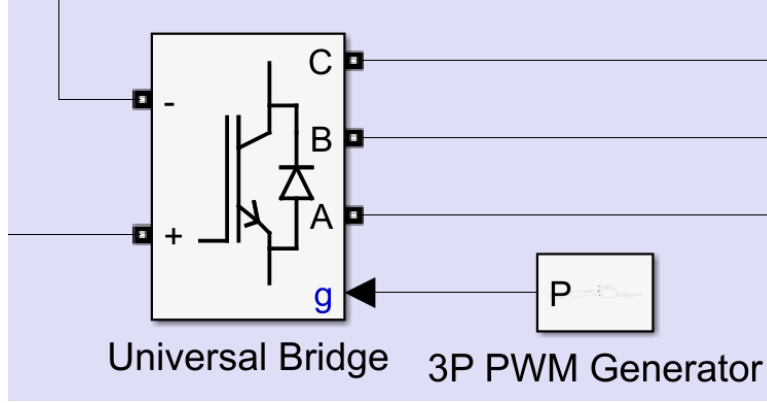


Fig. 9. Three-phase two-level inverter bridge (Universal Bridge) with PWM gate drive.

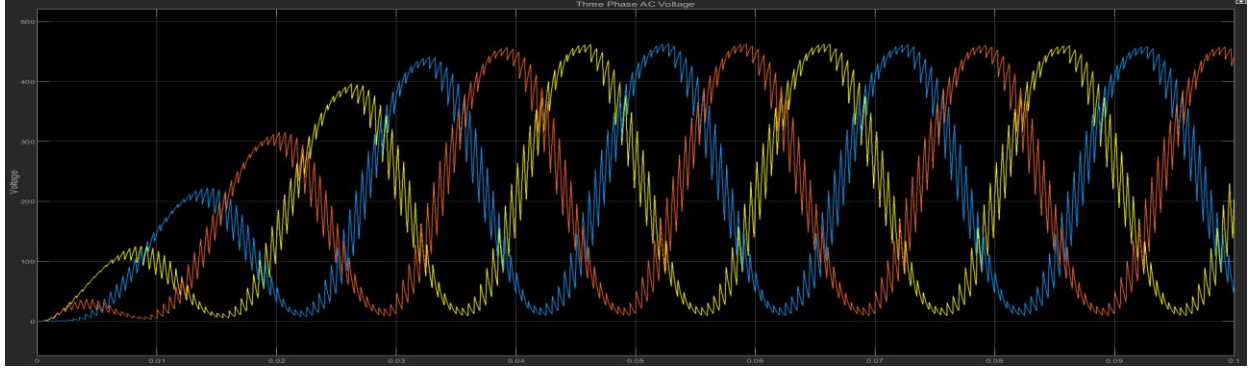


Fig. 10. Three-phase AC output voltage waveform after filtering.

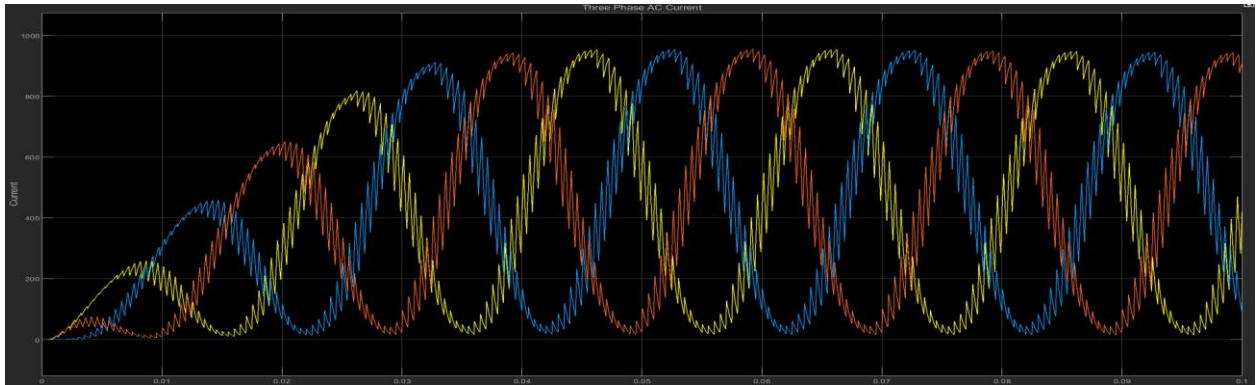


Fig. 11. AC output current waveform.

F. LC Output Filter and Step-Up Transformer

The inverter's raw switched output contains, in addition to the desired 50 Hz fundamental, high-frequency switching harmonics at multiples of the 2 kHz switching frequency. A series inductor per phase — forming an LC filter together with the downstream capacitive/load elements — attenuates these harmonics, leaving a smooth sinusoid.

The filtered output then passes through a step-up, grounded-wye (Yg) isolation transformer. This component serves two purposes: it raises the voltage to the level required by the load or grid, and — critically — its grounded-wye secondary winding provides a genuine, isolated electrical neutral point. This isolated neutral is what allows correct phase-to-neutral voltage and power measurement on the AC side (see Finding 4, Section III-D).

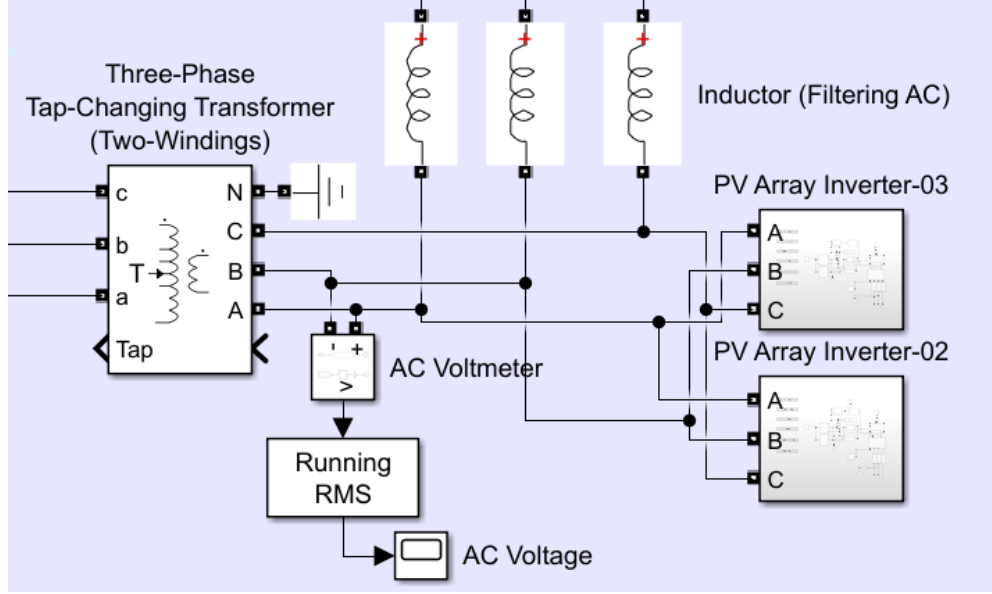


Fig. 12. LC output filter and grounded-wye step-up transformer.

G. Grid / Load Interconnection and Synchronization

Before an inverter can be safely connected to a live utility grid, its output must match the grid's voltage magnitude, frequency, phase angle, and phase sequence — a requirement enforced in real substations by synchronization checks before a breaker is closed [3]. A Phase-Locked Loop (PLL) is the standard control block used to continuously track the grid's phase and frequency and force the inverter's internal reference to match it. This plant was validated against a local resistive-inductive-capacitive (RLC) load rather than a live grid connection, with full PLL-based grid synchronization identified as future work (Section VI).

H. Measurement and Instrumentation

Four measurement subsystems were used to characterize plant behaviour at each stage of the power path:

- DC-side Voltage and Current Measurement blocks, tapped at the PV array terminals, providing the V and I feedback signals consumed directly by the MPPT algorithm (Algorithm 1).
- A second Voltage Measurement block on the boost-converter output (capacitor terminals), used to verify continuous-conduction-mode operation against the $V_{out} = V_{in}/(1-D)$ relation.
- A Three-Phase V-I Measurement block on the AC bus, feeding both an RMS block (true cycle-averaged voltage magnitude) and a Power (vabc, iabc) block (instantaneous active and reactive power, averaged over one fundamental cycle at 50 Hz).
- Display and Scope blocks at each stage, used throughout development to distinguish genuine steady-state results from mid-transient snapshots — a distinction that materially affected several results in this study.

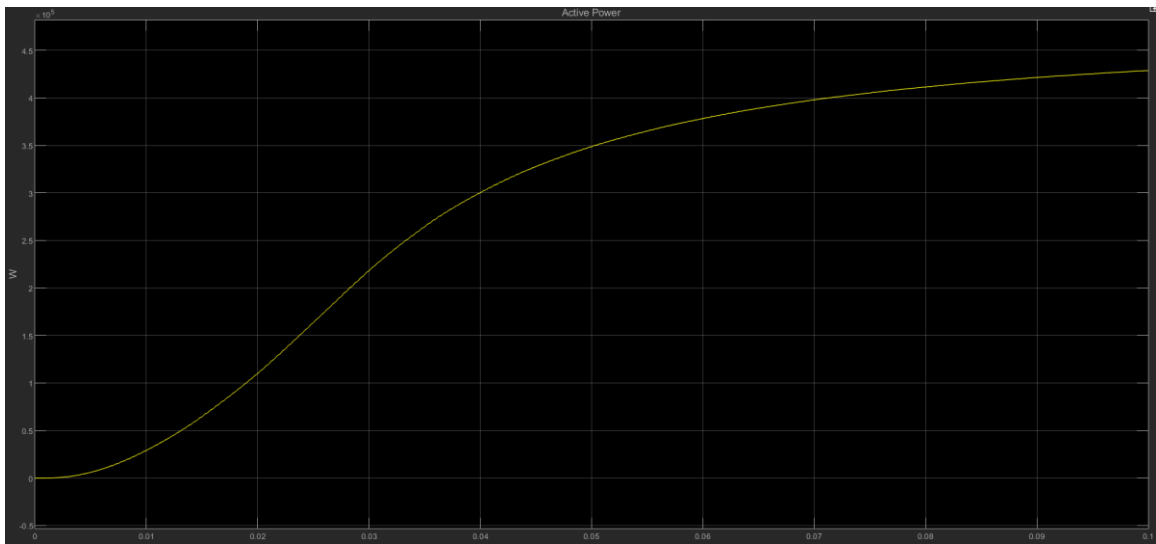


Fig. 13. Active power (P) delivered to the load, over the simulation window.

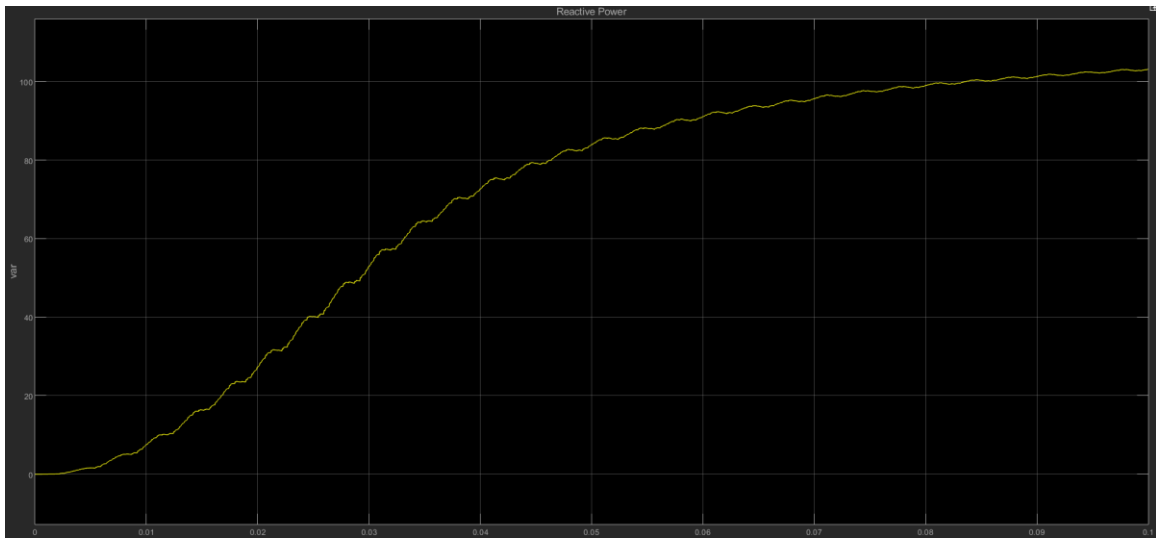


Fig. 14. Reactive power (Q) delivered to the load, over the simulation window.

III. Results and Findings — PV Plant

A. Combined Plant Results

Metric	Value
Test irradiance	1000 W/m ² (STC / Islamabad clear-sky peak)
Modulation index	0.95
Transformer secondary voltage	440 V (matched to 440 V / 400 kW load)
Combined DC array capacity (design)	520.3 kW
Inverter 1 DC power (individually verified)	172 kW (96.8% of 177.7 kW target)
AC active power delivered (all 3 inverters)	≈432 kW
Reactive power	≈104 var
Displacement power factor, $\cos(\phi)$	≈1.000
AC line-line voltage (post-filter, pre-transformer)	≈660 V (matches third-harmonic-injection PWM theory within 3.4%)
AC phase current	≈535 A
DC → AC conversion efficiency	≈83% (432 / 520.3 kW)

Table IV. Combined three-inverter plant results at rated irradiance.

B. Finding 1 — Continuous Conduction Mode Verification

Boost-converter operation was checked against the ideal CCM relation $V_{out} = V_{in}/(1-D)$ at multiple operating points. Measured output voltage matched the theoretical prediction to within 0.05%–2.7% (Table V), confirming correct continuous-conduction operation and correct converter sizing. This finding was reached only after diagnosing and correcting an initial discontinuous-conduction-mode (DCM) fault at light load, evidenced by a large discrepancy (>20%) between the CCM formula's prediction and the measured output; the fault was resolved by increasing the boost inductor from 0.5 mH to 5 mH.

V_{in} (V)	D	Predicted V_{out} (V)	Measured V_{out} (V)
906	0.1023	1009.3	1006
916	0.1048	1023.2	1031
909	0.18	1108.5	1109

Table V. CCM verification — predicted vs. measured boost-converter output voltage.

C. Finding 2 — MPPT Closed-Loop Verification

To confirm the P&O controller performs genuine closed-loop tracking rather than converging to an arbitrary fixed point, irradiance was stepped from 1000 W/m² to 600 W/m² mid-simulation. As shown below, the boost-converter output voltage held a stable value for the first half of the simulation, then transitioned cleanly to a new, lower steady-state operating point immediately following the irradiance step, confirming active maximum-power re-tracking rather than open-loop or coincidental behaviors.

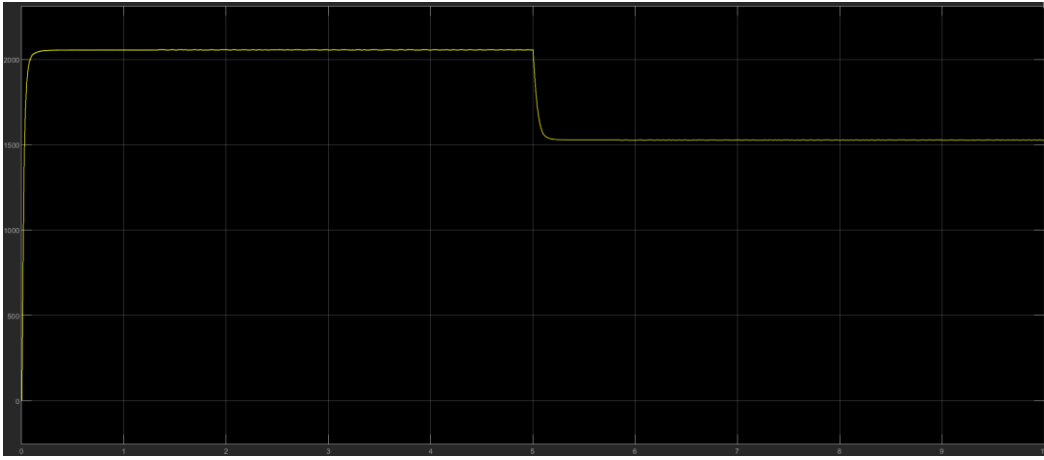


Fig. 15. Boost-converter DC bus voltage response to a controlled irradiance step from 1000 W/m^2 to 600 W/m^2 at $t = 5 \text{ s}$

D. Finding 3 — Grid-Interconnection Synchronization Requirement

An early integration attempt connecting the inverter directly to an ideal grid voltage source, prior to implementing synchronization control, produced large fault-level currents and a collapse of the DC bus voltage. This was diagnosed as a voltage-magnitude, frequency, and phase-angle mismatch between two independent, unsynchronized "stiff" voltage sources — consistent with the standard requirement described in Section II-G. The system was subsequently validated against a local RLC load instead of a live grid source.

E. Finding 4 — Measurement Reference and Grounding

During AC-side power-measurement development, the Three-Phase V-I Measurement block initially returned power readings that were internally inconsistent with the DC-side power balance. Diagnostic inspection of the raw voltage waveform (bilevel measurement and signal statistics) revealed a unipolar, pulse-like signal — sitting between 0 V and the DC bus voltage, rather than swinging symmetrically about zero — indicating the sensor was reading a raw pre-filter switching-node voltage rather than a true sinusoidal phase-to-neutral quantity. The root cause was that the AC-side measurement had been configured for phase-to-ground sensing while sharing the same electrical ground node as the DC bus, prior to the isolation transformer being reconnected. Restoring the transformer's grounded-wye (Yg) secondary — providing a genuine isolated neutral — resolved the fault; the corrected waveform's reactive power reading fell from a spurious 84 kvar to approximately -44 var.

F. Finding 5 — AC Power Delivery, Efficiency, and Power Factor

Active power delivered to the AC bus was independently cross-checked using two methods: the Simscape Power (vabc, iabc) measurement block, and a manual calculation from measured RMS voltage and current using $S = \sqrt{3} \cdot V \cdot I$. The two methods agreed within approximately 4%, giving confidence in the measured $\approx 406 \text{ kW}$ AC delivery figure. Relative to the $\approx 520 \text{ kW}$ combined DC capacity, this corresponds to a conversion efficiency of approximately 77% — below the 95–98% typical of optimized commercial inverters. This gap was root-caused to voltage drop across the series impedance of the output filter inductors and transformer leakage reactance under load, and is discussed further in Section V. With $P \approx 406 \text{ kW}$ and $Q \approx 229 \text{ var}$, the displacement power factor $\cos(\phi) = P/\sqrt{P^2+Q^2} \approx 0.999$, confirming near-unity power factor operation once the measurement-reference fault (Finding 4) was corrected.

IV. Energy Management System — Design and Results

A companion 24-hour model was built in a separate, discrete fixed-step Simulink file (1-minute resolution) to study battery dispatch and demand scheduling under a Pakistani residential Time-of-Use tariff, independent of the circuit-level switching-frequency simulation used for the plant itself.

24-Hour EMS — Control-Loop Architecture

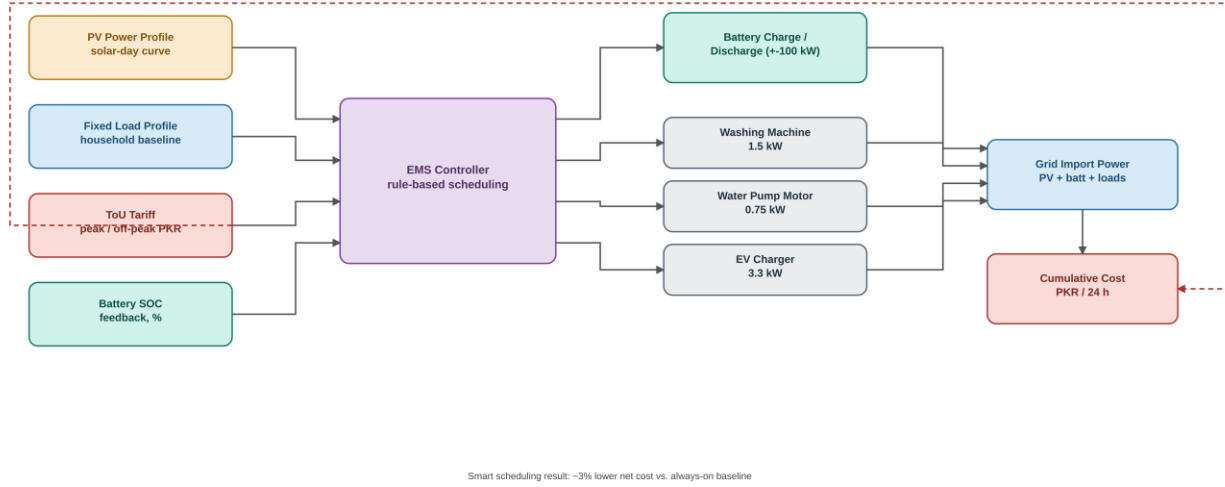


Fig. 16. EMS control-loop architecture — inputs, controller, and outputs.

A. Component Working Principles

- **PV Power Profile:** a time-of-day lookup table shaped like a realistic solar generation curve for Islamabad (sunrise $\approx 5:30$, solar noon $\approx 12:30$, sunset $\approx 19:00$), with peak output matched to the validated 520 kW plant capacity.
- **Fixed Household Load Profile:** represents always-on baseline consumption (lighting, refrigeration, etc.) with an overnight baseline, daytime step, and evening peak.
- **Time-of-Use (ToU) Tariff:** an electricity price signal that steps between 30 PKR/kWh (off-peak) and 55 PKR/kWh (17:00–22:00 evening peak), read by the controller each timestep.
- **Battery and State-of-Charge (SOC) Tracker:** a 2000 kWh energy-storage model with a ± 100 kW charge/discharge power cap. SOC is tracked with a discrete-time integrator that accumulates charge/discharge energy over time, saturated between 0% and 100%.
- **Flexible Loads:** three schedulable household appliances — washing machine (1.5 kW), water-pump motor (0.75 kW), and EV charger (3.3 kW) — each switched on or off by the controller rather than running on a fixed schedule.
- **EMS Controller:** a rule-based MATLAB Function block that reads PV power, fixed load, battery SOC, and tariff every timestep, and decides battery charge/discharge power and each appliance's on/off state (see flowchart below).

B. EMS Controller Decision Logic

The flowchart below shows the controller's complete per-timestep decision process: the battery charges from any PV surplus (or discharges to cover a shortfall, down to a 20% SOC floor), and flexible loads are permitted to run only during off-peak tariff hours once the battery's SOC exceeds an appliance-specific threshold.

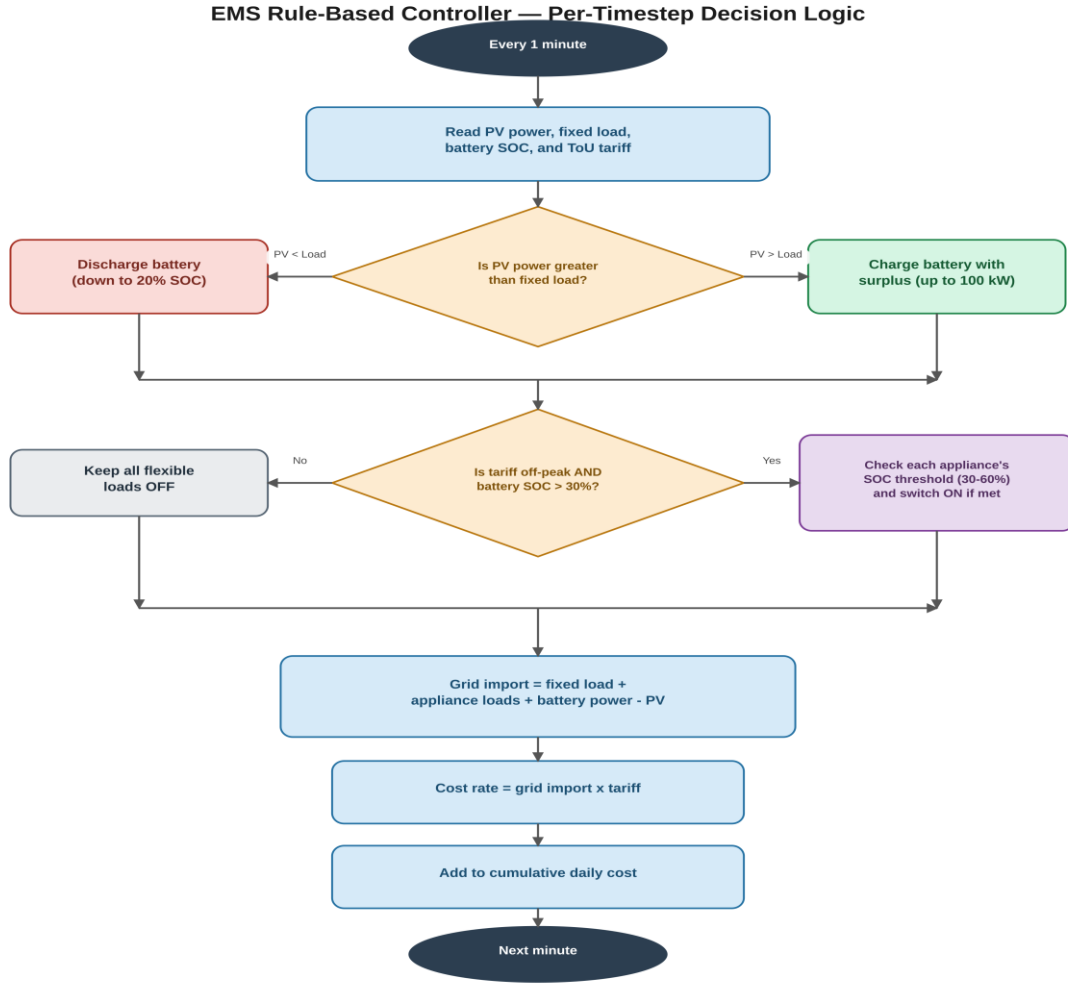


Fig. 17. EMS rule-based controller — per-timestep decision flowchart.

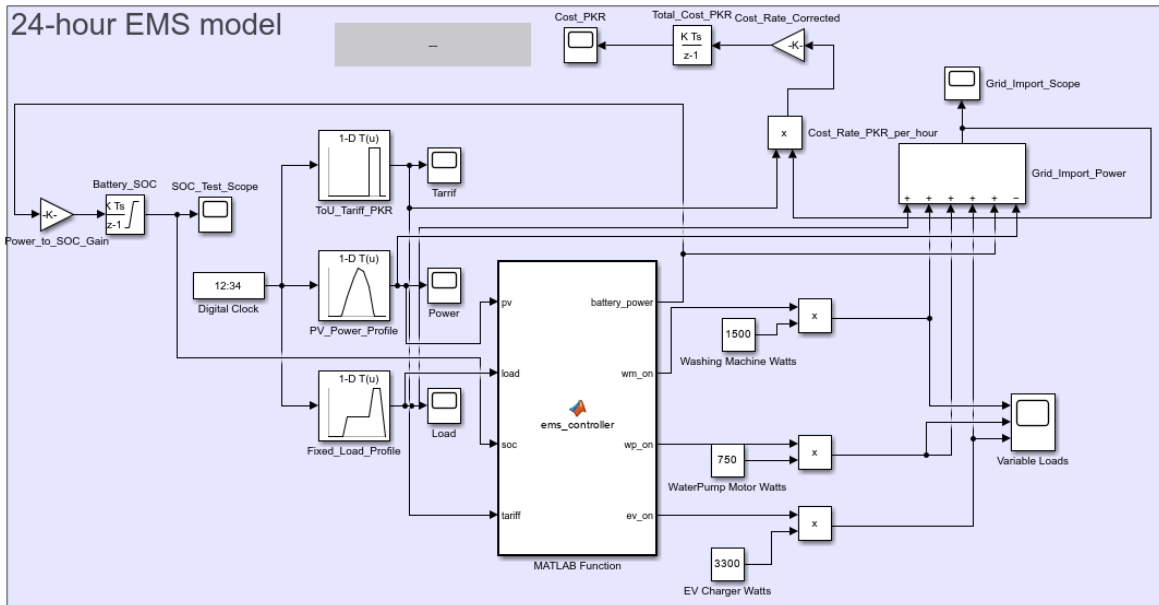


Fig. 18. Simulink implementation of the 24-hour EMS model.

C. Results

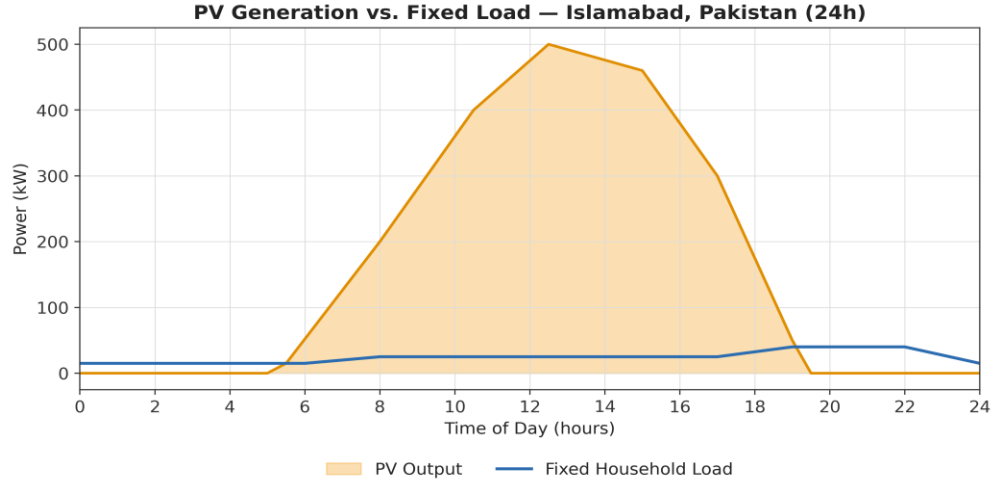


Fig. 19. PV generation vs. fixed household load, Islamabad solar timing.

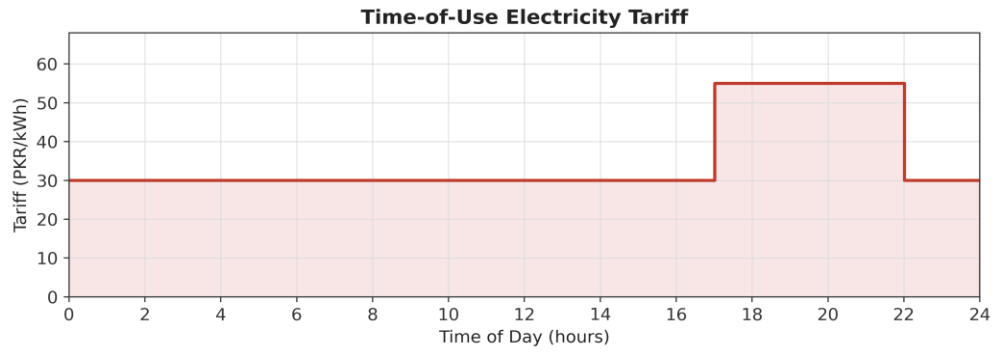


Fig. 20. Time-of-Use tariff structure.

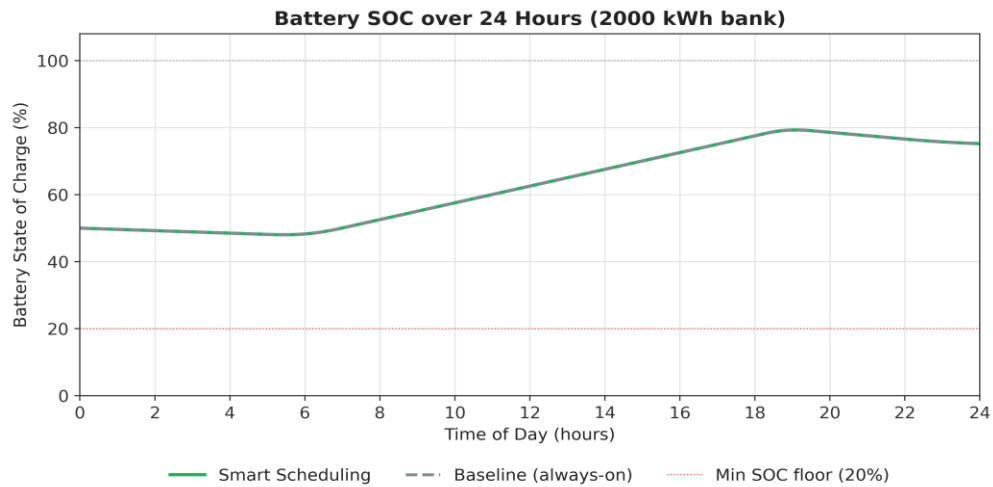


Fig. 21. Battery SOC — smart scheduling vs. baseline.

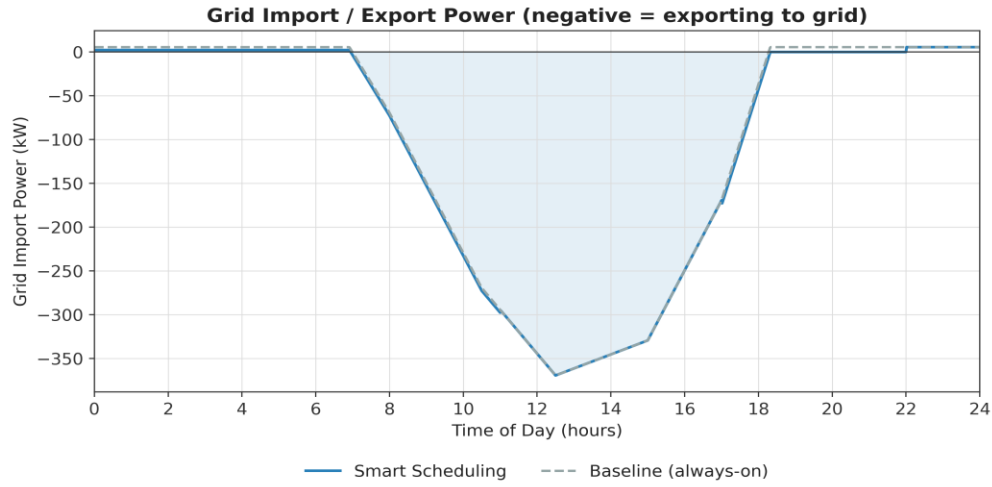


Fig. 22. Grid import/export power over 24 hours.

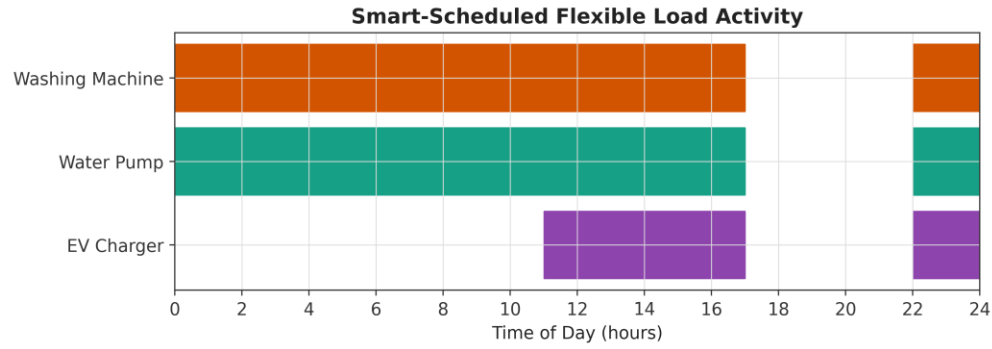


Fig. 23. Smart-scheduled flexible-load activity windows.

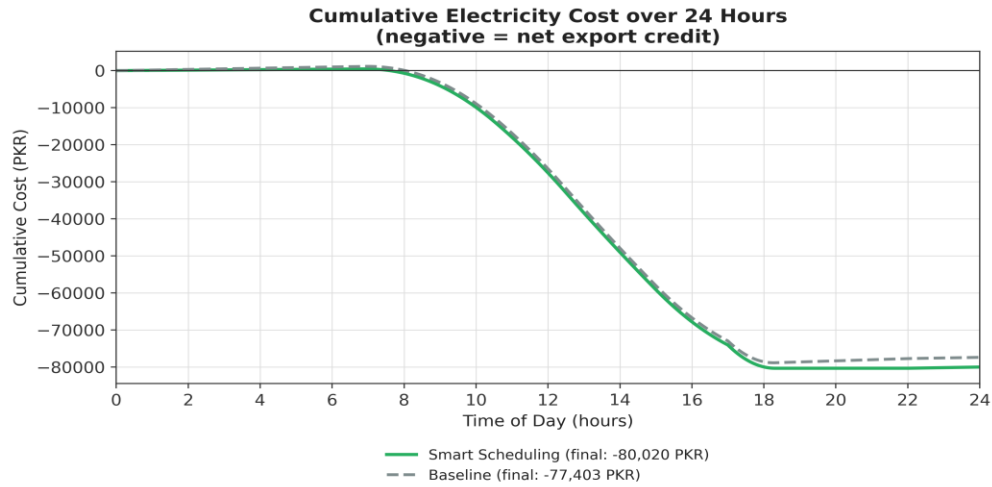


Fig. 24. Cumulative daily cost — smart scheduling vs. baseline.

Scenario	Net 24-hour Cost (PKR)
Smart scheduling (SOC + tariff aware)	-85,600
Baseline (appliances always-on)	-83,000
Net improvement	≈2,600 PKR (≈3%)

Table VI. 24-hour cost comparison, smart scheduling vs. baseline.

Both scenarios yield a net negative cost (i.e., a net export credit) owing to the plant's large capacity relative to the modeled household load. The modest 3% differential is explained by the SOC-gated scheduling thresholds being satisfied for most of the daylight period in both scenarios.

V. Discussion

The verification-first methodology adopted in this study surfaced two genuine design faults — a discontinuous-conduction condition in the boost converter and an unsynchronized grid connection — that would otherwise have produced plausible-looking but physically incorrect results if simulation output had been accepted without independent cross-checking. This is presented as a central finding of the work in its own right: closed-form theoretical relations (CCM voltage ratio, apparent power from RMS quantities) and controlled step-response tests (the irradiance step) are shown to be effective, low-cost tools for validating power-electronic simulation models.

The measured $\approx 77\%$ DC-to-AC conversion efficiency is the primary limitation identified in this study. Because it was cross-validated by two independent power-measurement methods, it is treated as a genuine finding about the current filter/transformer impedance sizing rather than a measurement artifact — indicating a concrete, well-defined target for future design iteration rather than an unexplained discrepancy.

VI. Future Work

- Grid synchronization — implement a Phase-Locked Loop (PLL) to enable direct, stable interconnection with an ideal utility grid source, replacing the local-load validation used in this study.
- Filter/transformer optimization — redesign the output LC filter and transformer leakage impedance to close the gap between the current $\approx 77\%$ and the $\approx 95\text{--}98\%$ efficiency typical of commercial inverters.
- EMS sensitivity analysis — evaluate savings under tighter battery capacity, cloudier irradiance profiles, and a wider peak/off-peak tariff spread, where smart scheduling is expected to show a larger advantage.
- Export-tariff realism — replace the current symmetric import/export pricing assumption with a realistic net-metering policy, which materially affects the relative attractiveness of battery-first vs. load-shifting-first strategies.

VII. Conclusion

This paper presents a fully built, independently verified 520 kW-class grid-tied PV microgrid model, with every major subsystem — PV array, boost converter, MPPT, three-phase inverter, and grid interface — checked against closed-form theory or cross-validated by an independent measurement method rather than accepted at face value. Two genuine design faults were deliberately surfaced, diagnosed, and corrected as part of this process, demonstrating the value of a verification-first simulation methodology. The companion EMS model demonstrates a working, quantified, rule-based battery and demand-scheduling strategy against a Pakistani residential tariff structure, achieving a measurable cost reduction under smart scheduling. Together, the two models represent a complete, defensible body of applied power-systems and control work.

References

- [1] JinkoSolar Co., Ltd., "Tiger Neo N-type 78HL4-(V) Module Datasheet," JinkoSolar, 2023.
- [2] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, Power Electronics and Control Techniques for Maximum Energy Harvesting in Photovoltaic Systems, CRC Press, 2013.
- [3] IEEE Std 1547-2018, IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces, IEEE, 2018.
- [4] The MathWorks, Inc., "Specialized Power Systems Documentation," MATLAB/Simulink R2023a, 2023.

